Closing the cubic singlet two-point spectrum
A charged covariance lift for the linear residual

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Local research note — 6 November 2026

Abstract

The remaining linear–quadratic and linear–cubic cross spectra can be tested using the existing embedded-ensemble covariance bridge. The necessary extra step is a charge-dependent lift of a $(p-1)$-body ensemble to a p-body operator. We give its exact covariance and marked-product identities. A single centered filter then cancels in every fixed-degree mixed moment. Together with the preceding cubic–quadratic cancellation, this closes the positive two-energy measure of the conserved-subtracted cubic derivative insertion. Its small-coupling normalized time correlation is a definite combination of sech^6 and sech^4, rather than the isolated cubic-primary kernel.

1 The completed two-point object

Use standard CAR, a complex Gaussian p-body Hamiltonian $H = H_s$ with overlapping index sets included, and the annealed normalized trace at $\beta = \mu_{\mathrm{chem}} = 0$. Set $\hbar = 1$, $L = [H, \cdot]$, $\mathsf{J}_H B = \{H, B\}/2$, and first take

$$N, p \to \infty, \qquad \lambda = p^2/N > 0, \qquad q = e^{-4\lambda}, \qquad \sigma^2 = \mathcal{J}_s^2/\lambda.$$

The preceding notes [1] constructed the linear residual F, quadratic singlet Y, and cubic residual R. Their diagonal two-energy measures have the known Mehler weights q, q^2, q^3. We prove the remaining off-diagonal statements

$$\boxed{d\nu_{FY} = d\nu_{FR} = 0 \quad \text{in DS.}} \tag{1}$$

Along with the previous $d\nu_{RY} = 0$, these diagonalize the *two-point* spectral matrix. They do not imply independence of the operators at higher point order.

For $M_3 = \sum_i c_i^\dagger L^3 c_i$, subtract the conserved polynomial P_N defined below. Its complete limiting measure is

$$\boxed{d\nu_{UU^\dagger} = V_F\, d\mu_q + V_R\, d\mu_{q^3} + W\left[a\frac{E+E'}{2} + \frac{3}{2}(E-E')\right]^2 d\mu_{q^2},} \tag{2}$$

where $U = M_3 - P_N$, each μ_r has unit mass, and a, V_F, V_R, W are specified in Sec. 5. The formula retains the non-Hermitian time descendant of the actual insertion. It is positive and fixes all its two-energy moments and every fixed-time two-point function.

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2 An operator L^2 bridge, not only a covariance estimate

In charge sector m let $H = \sum_\nu H_\nu$ be the irreducible rank decomposition, $C = \langle H^2\rangle_m$, and $\langle H_\nu^2\rangle_m = C\omega_\nu^{(p)}$. Put $h = H/\sqrt{C}$, $V_{F,N} = \langle F^2\rangle_m$, $y = F/\sqrt{V_{F,N}}$. The preceding rank formulas give, with $s = p - \nu$,

$$R_1(s) = \frac{\omega_{p-s}^{(p-1)}}{\omega_{p-s}^{(p)}} = \frac{s(N-2p+s+1)(N-m+p)}{(m-p+1)p^2}, \tag{3}$$

$$\widehat{y} = \sqrt{\lambda}\sum_\nu [R_1(p-\nu) - 1]h_\nu, \qquad \|y - \widehat{y}\|_2 \longrightarrow 0. \tag{4}$$

Indeed s tends to Pois(λ), the normalized F filter tends to $(s-\lambda)/\sqrt{\lambda}$, and $R_1(s) \to s/\lambda$. The factorial rank tail and polynomial bounds in the earlier notes yield weighted L^2, not merely L^1, convergence. Also $\sum \omega^{(p)} R_1 = 1$ exactly.

For one $\widehat{y}$ slot, its Wick edge with an h slot has covariance $\sqrt{\lambda}(\mathsf{K}_{p-1} - \mathsf{K}_p)$, with K_r the unit-variance r-body covariance at charge m. Ordinary words in h need no more information. A derivative vertex, however, also samples H_{m-1}, and its covariance must be lifted consistently before the two terms are compared.

Let $V^{(r)}$ be an independent embedded ensemble, normalized to unit variance at the reference charge m. For $r = p - 1$ define

$$\widetilde{V}^{(p-1)} = g(Q_f)V^{(p-1)}, \qquad g(n) = \frac{n-p+1}{m-p+1}. \tag{5}$$

Multiplication by $Q_f - (p-1)$ turns a normal-ordered $(p-1)$-body monomial into a p-body operator. More explicitly a rank-ν operator of body order r carries charge factor $\binom{n-\nu}{r-\nu}$. The identity

$$g(n)\frac{\binom{n-\nu}{p-1-\nu}}{\binom{m-\nu}{p-1-\nu}} = \frac{\binom{n-\nu}{p-\nu}}{\binom{m-\nu}{p-\nu}} \tag{6}$$

shows that the lifted covariance, with one endpoint at m and one at any allowed n, is exactly $R_1(p-\nu)$ times the normalized p-body covariance in each rank channel. Thus (5) extends the signed edge replacement to the charged insertion; it is not an approximation that ignores the adjacent sector.

3 Why the derivative mark survives the lift unchanged

Write $\mathfrak{D}B = Q_f B - \sum_i c_i^\dagger B c_i$. For any neutral B and scalar function f,

$$(\mathfrak{D}[f(Q_f)B])_m = f(m-1)(\mathfrak{D}B)_m + m[f(m) - f(m-1)]B_m. \tag{7}$$

Consider a product of $k = 2$ or 3 factors with exactly one factor of body order $p-1$, the others of order p. Call the unlifted product B, and set $\eta = g(m-1) = (m-p)/(m-p+1)$. Then

$$\boxed{[(kp - \mathfrak{D})(gB)]_m = \eta[(kp - 1 - \mathfrak{D})B]_m + \frac{(k-1)p}{m-p+1} B_m.} \tag{8}$$

The first term is the natural normal-order contraction count for the lower-body product. The additive correction is $O(p/N) = O(1/p)$ in the central charge window, and $1 - \eta = O(1/N)$.

This is enough for our vertices. For V_1, V_2, V_3 of body order p,

$$K_{12} = [(2p - \mathfrak{D})(V_1 V_2)]V_3, \tag{9}$$
$$K_{23} = V_1[(2p - \mathfrak{D})(V_2 V_3)], \tag{10}$$
$$K_{13} = (3p - \mathfrak{D})(V_1 V_2 V_3) - K_{12} - K_{23}. \tag{11}$$

The quadratic vertex is $A = (2p - \mathfrak{D})H^2$, and $T_3 = K_{13} - (K_{12} + K_{23})/2$. The residual R is a fixed linear combination of these single marks and the unmarked cubic word, with internal coupling contractions deleted by $\mathcal{W}_3$. If the shifted endpoint is outside the vertex no mark changes; if inside, (8) treats it, including the nonadjacent mark.

The remaining large-N input is the existing fixed-length mixed-body chord limit [2, 3], with the local determinant mark used in our earlier notes. For a fixed allowed pairing, each index-set size is p or $p-1$. Its pair-overlap mean $r_i r_j / N \to \lambda$, while the probability of a triple overlap is $O(p^3/N^2) \to 0$. The finite overlap moment-generating functions are bounded in a complex neighborhood of zero by sampling-without- replacement exponential-moment bounds. Consequently the marked generating functions converge locally uniformly; taking their fixed mark derivatives commutes with the limit. The limiting marked diagram therefore does not change under this one-body shift.

This also bounds each unmarked and marked term in (8), so its $O(1/p)$ correction really vanishes. The argument is uniform on $|m - N/2| \le N^{2/3}$; polynomial bounds and the exponentially small charge tail extend it to the full trace. No number of external Hamiltonians is allowed to grow with N in this argument.

4 Cancellation in all fixed moments and in the spectral measure

Take $B = Y_N$ or R_N, and fixed nonnegative u, v. First replacing y by $\widehat{y}$ in $\langle y H^u B H^v \rangle$ is legitimate directly at the operator level:

$$|\langle (y - \widehat{y}) H^u B H^v \rangle| \le \|y - \widehat{y}\|_2 \, \langle B H^{2u} B H^{2v} \rangle^{1/2} \longrightarrow 0. \tag{12}$$

The second factor has a finite limit by the already established diagonal two-energy kernels. In Y_N, ensemble centering deletes the internal quadratic contraction; in R_N, $\mathcal{W}_3$ deletes all internal cubic contractions before the signed replacement.

There is exactly one filtered slot. In every remaining Wick pairing it is paired with one of the ordinary Hamiltonian slots. Equations (3)–(6) replace this edge by a lifted $(p-1)$-body edge minus a p-body edge, with coefficient $\sqrt{\lambda}$. Section 3 proves their individual diagram limits are identical, including when the endpoint lies inside a derivative vertex. Their difference is zero. There are finitely many pairings at every fixed u, v. Therefore

$$\langle F H^u Y H^v \rangle_{\mathrm{DS}} = \langle F H^u R H^v \rangle_{\mathrm{DS}} = 0 \quad \text{for all fixed } u, v. \tag{13}$$

This uses neither finite-size independence nor the false assumption that time evolution preserves ordinary Gaussian chaos.

For Hermitian insertions define $d\nu_{AB} = \mathbb{E}\,\mathrm{tr}[P(dE) A P(dE') B]$. The matrix of such measures is positive, so spectral Cauchy–Schwarz bounds the total variation and polynomial tails of its off-diagonal entries by the diagonal measures. The latter converge to compactly supported Mehler measures; their higher moments give uniform integrability. Every subsequential cross-measure limit thus has compact support and all polynomial moments zero by (13). Polynomial density on the compact energy square makes it the zero measure. This proves (1) and fixed-time cross-correlation convergence.

For orientation, the first energy-sum overlap is not zero at finite size. The preceding exact counterterms give

$$\langle F \mathsf{J}_H Y_N \rangle = \langle F G_N \rangle, \qquad p \frac{\langle F \mathsf{J}_H Y_N \rangle}{\langle F^2 \rangle} \longrightarrow \frac{1 - 2\lambda - c}{2}, \quad c = \frac{2\lambda(1-q)}{1+q}. \tag{14}$$

It supplies a nontrivial finite-size check of the vanishing unscaled cross spectrum, rather than an identically zero test case.

5 The positive measure of the actual cubic insertion

Recall $A = \sum_i (Lc_i)^\dagger Lc_i$, $T_3 = (M_3 + M_3^\dagger)/2$, and $(T, S, Z) = \mathcal{W}_3(T_3, \mathsf{J}_H A, H^3)$, with

$$R = T - aS - bZ, \quad a = -\frac{(1-q)^2}{1+q^2}, \quad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}.$$

The preceding exact finite-N reduction is

$$P_N = (ac_N + b)H^3 + [\bar{t} - (ac_N + b)\bar{z}]H, \tag{15}$$

$$U_N := M_3 - P_N = R + a\mathsf{J}_H Y_N + F + \tfrac{3}{2}LY_N - aB_S - bB_Z. \tag{16}$$

Here $Y_N = A - a_0 - c_N(H^2 - C)$ uses its exact sector projection, and $\|B_S\|_2, \|B_Z\|_2 = O(1/p)$. All scalar coefficients are deterministic functions of charge. The subtraction is an explicit conserved polynomial, not a claim of projection onto every finite-size conserved operator.

Let $d\mu_r = \rho_q(E)\rho_q(E')K_r(E, E')\,dE\,dE'$ denote the existing unit-mass Mehler measure [3, 2]. Its support is $|E|, |E'| \le 2\sigma/\sqrt{1-q}$, and its normalized polynomial expansion is $K_r = \sum_{n\ge 0} r^n p_n(E)p_n(E')$. Define

$$V_F = 64\mathcal{J}_s^6 q^2, \qquad W = 4\mathcal{J}_s^4[\lambda^{-1} + 4q/(1+q)], \tag{17}$$

$$V_R = \sigma^6 \frac{\lambda A_1 + \lambda^2 A_2}{D^2}, \qquad D = (1+q^2)(1+q+q^2), \tag{18}$$

$$A_1 = 4(1-q^2)(1+q^2)(1+q+q^2)^2, \tag{19}$$

$$A_2 = 16q^2(1+q)(1+q^2)(1+q+q^2)(2q^2 - q + 2). \tag{20}$$

The three off-diagonal measures now vanish, and the small linear remainder in (16) vanishes in fixed two-energy moments and fixed-time L^2 correlators. This proves (2), where for the non-Hermitian U the second insertion is explicitly $U^\dagger$.

In particular, put $r = q^2$. Its norm is

$$\mathcal{N} = V_F + V_R + W\sigma^2 \left[\frac{a^2(1+r)}{2} + \frac{9}{2}(1-r)\right]. \tag{21}$$

The mixed $(E + E')(E - E')$ term in (2) integrates to zero at every fixed frequency, by $(E, E') \mapsto (-E', -E)$ symmetry. It remains present in the full two-energy measure. For fixed $\lambda > 0$ that measure is absolutely continuous and has no zero-frequency atom. The unsubtracted polynomial in M_3 must still be kept if one studies its full static component.

6 Small coupling: a dominant time descendant

Write $C_d(t) = \int e^{it(E-E')} d\mu_{q^d}$. The known small-coupling kernels obey $C_d \to \operatorname{sech}^{2d}(2\mathcal{J}_s t)$. The frequency-weighted Y limit is justified by convergence of the higher frequency moments, so that $-C_2''$ may be used here; this is stronger than just differentiating pointwise convergence. As $\lambda \to 0^+$, after DS,

$$V_R \sim \frac{32\mathcal{J}_s^6}{\lambda}, \quad V_F \sim 64\mathcal{J}_s^6, \quad W \sim \frac{4\mathcal{J}_s^4}{\lambda}, \quad a = -8\lambda^2 + O(\lambda^3).$$

The energy-sum contribution is negligible: its absolute correlation is bounded by its equal-time norm $a^2 W \sigma^2 (1+q^2)/2 = O(\lambda^2 \mathcal{J}_s^6)$. The cubic residual and time descendant instead contribute

$$\mathcal{N} \sim 176\mathcal{J}_s^6/\lambda, \tag{22}$$

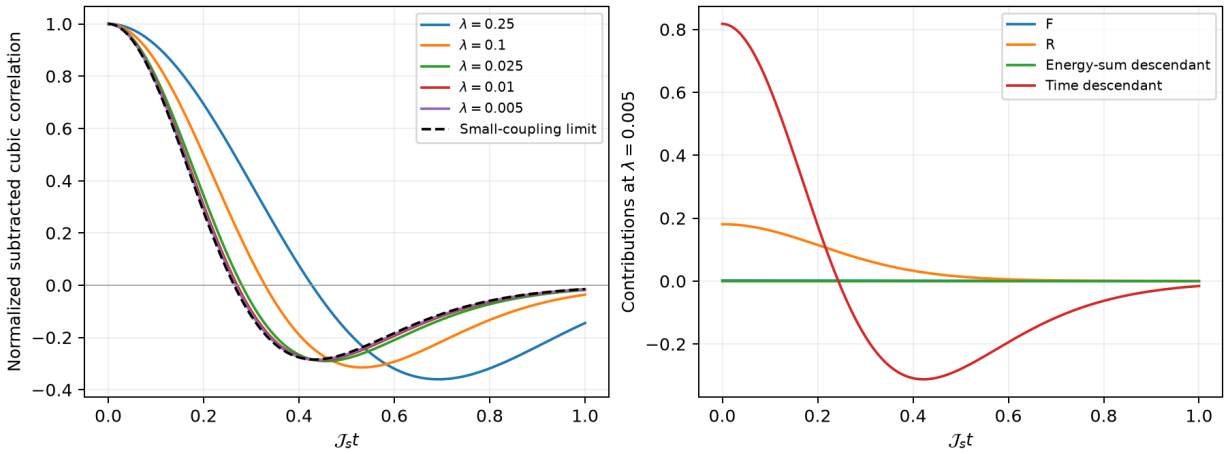
$$\boxed{\frac{\langle U(t)U^\dagger\rangle_{\mathrm{DS}}}{\mathcal{N}} \longrightarrow \frac{47\operatorname{sech}^6(2\mathcal{J}_s t) - 36\operatorname{sech}^4(2\mathcal{J}_s t)}{11}.} \tag{23}$$

The fractions of the leading equal-time norm are $2/11$ from R and $9/11$ from $\frac{3}{2}LY$. Its first positive zero is $\mathcal{J}_s t = \frac{1}{2}\operatorname{arcosh}\sqrt{47/36} \simeq 0.263953$; the initial normalized frequency moment is $552\mathcal{J}_s^2/11$.

A negative time-correlation interval is consistent with a positive frequency density. With $C(t) = \int S(\omega) e^{it\omega} d\omega/(2\pi)$, the limiting normalized density is explicitly

$$S_U(\omega) = \frac{47(\omega/\mathcal{J}_s)^2 + 128}{880} S_2(\omega) \geq 0, \tag{24}$$

$$S_2(\omega) = \frac{\pi\omega(\omega^2 + 16\mathcal{J}_s^2)}{96\mathcal{J}_s^4 \sinh(\pi\omega/(4\mathcal{J}_s))}, \qquad S_U(0) = \frac{16}{165\mathcal{J}_s}. \tag{25}$$



The left panel shows the full normalized subtracted correlation; the right resolves its four diagonal contributions. The original derivative insertion therefore has a substantially different two-point shape from the isolated q^3 residual.

7 Validation, attribution and the next physical step

The charged bridge is checked independently of rank decomposition by the explicit CAR Casimir

$$\mathcal{C}(B) = \tfrac{1}{2}\sum_{ij}[c_i^\dagger c_j, [c_j^\dagger c_i, B]], \qquad \mathcal{C}(H_\nu) = \nu(N+1-\nu)H_\nu.$$

Equation (3) is therefore a quadratic-rank polynomial map on the full finite matrix. Three systems $(N,p,m) = (4,2,2),(5,2,3),(6,2,3)$ give eight charged covariance blocks, totaling 288772 entrywise equalities. Integer cross multiplication verifies them; 15 exact marked-product identities test (8). Twelve full-rank sums at 80/120 digits check (4) and the nonzero $1/p$ leakage (14), with all finite-size deviations retained.

The five couplings use Jacobi sizes 35, 55, 142, 309 and 582 for $\lambda = 0.25, 0.1, 0.025, 0.01, 0.005$ over $0 \le \mathcal{J}_s t \le 1$. Each is checked with 16 extra rows. An independent positive Mehler quadrature at 128/192 nodes checks the first two couplings, both time values and the norm/second frequency moment; its largest time-value difference from Jacobi is 3.57×10^{-14}. At $\lambda = 0.005$ the maximum absolute deviation from (23) is still 0.0131. Relative errors are inappropriate across the correlation's zeros; they are not silently reported as small.

For reproducibility the analytic Jacobi truncation bound also covers the energy weights. Set $B = 2\sigma/\sqrt{1-q}$, $\delta_n = 2\sum_{k\ge n-2}(B|t|)^k/k!$. Propagation of e_0, Je_0, J^2e_0 differs from its n-row version by at most $\delta_n, B\delta_n, B^2\delta_n$. The normalized correlation error is bounded by

$$\frac{\delta_n}{\mathcal{N}}[2(V_F+V_R) + WB^2(2a^2+18)] < 10^{-12}.$$

Floating-point roundoff and quadrature refinement are empirical checks, not interval-certified error bounds. Independent chord enumeration checks nine moments through total degree four and the exact Laurent coefficients 176 and 8832. Four Fourier integrals at 50/80 digits check the positive density in (24), retaining the cancellation between its two time kernels.

This closes a specific microscopic two-point problem. The next physical task is the mixed F, Y, R higher-point structure and its spatial/bulk interpretation. Vanishing pairwise cross spectra does not fix mixed three/four-point functions, locality or particle masses. The known covariance, chord and Mehler machinery is used as input; we do not claim a new solution of those standard kernels.

References

[1] A. Day and Chengheng, preceding local notes: *The linear cubic residual has a full q kernel*; *Reducing the raw cubic singlet*; quadratic and cubic full-kernel notes. See `manuscript/susskind-linear-kernel`, `susskind-cubic-cross`, `susskind-singlet-kernel`, `susskind-cubic-kernel`.

[2] M. Berkooz, P. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Secs. 3.2–3.3 and 7.

[3] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 2.1, Eqs. (2.17)–(2.21), and Sec. 4.1.